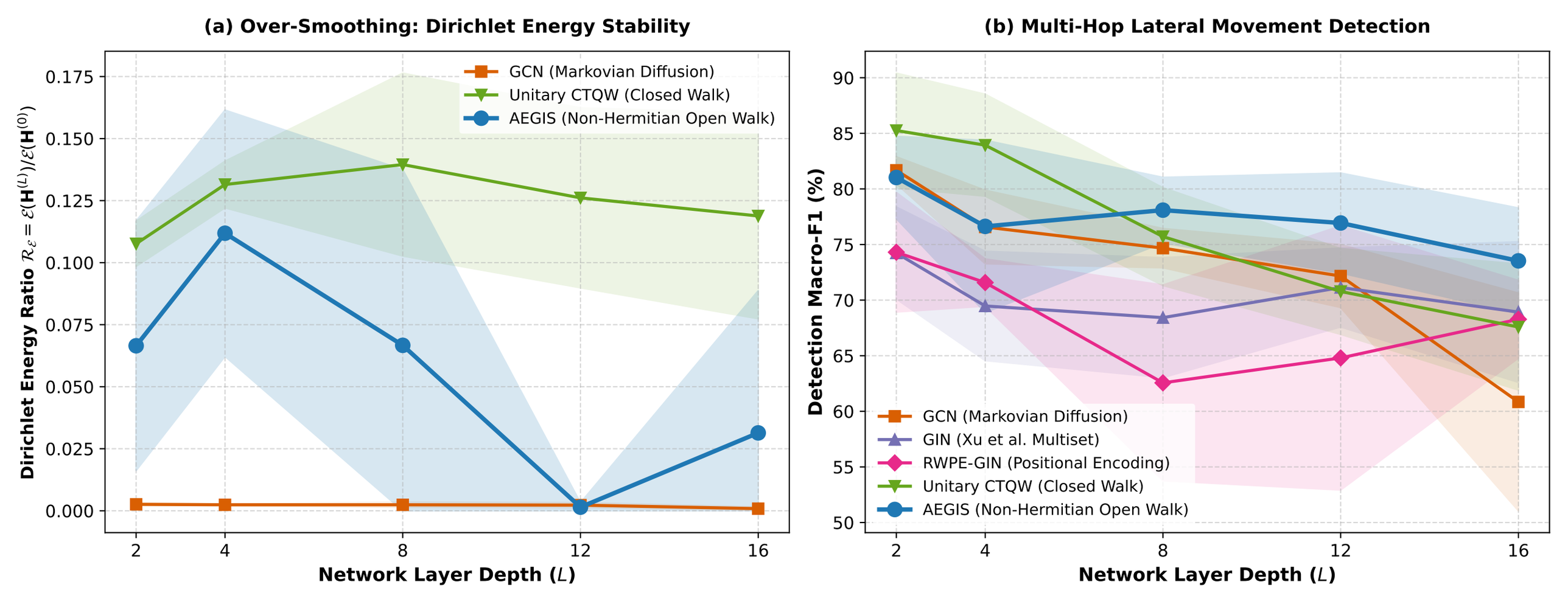
(a) Over-Smoothing: Dirichlet Energy Stability
Dirichlet Energy Ratio $\mathcal{R}_\mathcal{E} = \mathcal{E}(\mathbf{H}^{(L)})/\mathcal{E}(\mathbf{H}^{(0)})$
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
2
4
8
12
16
Network Layer Depth (L)
GCN (Markovian Diffusion)
Unitary CTQW (Closed Walk)
AEGIS (Non-Hermitian Open Walk)
(b) Multi-Hop Lateral Movement Detection
Detection Macro-F1 (%)
90
85
80
75
70
65
60
55
50
2
4
8
12
16
Network Layer Depth (L)
GCN (Markovian Diffusion)
GIN (Xu et al. Multiset)
RWPE-GIN (Positional Encoding)
Unitary CTQW (Closed Walk)
AEGIS (Non-Hermitian Open Walk)